

OPERATOR'S GUIDE



U.S.A.I Document 770010

Revision 2

GE DOCUMENT PART NUMBER: 2377425-4

**GE Signa® 1.5T
8-CHANNEL BODY ARRAY**

GE Catalog Part Number: M3087JE

USAI Part Number: 160183

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SECTION 1 – INTRODUCTION

1-1 Operator's Guide Information

The operator's guide provides information on how to use the 8-Channel Body Array. The user should read this guide before using the coil for imaging. The user is advised to refer to the MR scanner manual in conjunction with this guide.

Direct any questions regarding this manual to:

USA Instruments, Inc.
1515 Danner Drive
Aurora, Ohio 44202

1-2 Use of Symbols

Coil users should refer to the operator's guide for caution and warning explanations, using the symbols defined below.



Cautions provide information on possible equipment hazards.



Warnings provide information on possible hazards to human beings.

1-3 Coil Classification

Class II equipment (the symbol is shown right).



Type BF applied part (the symbol is shown right).



Ordinary equipment.

Suitable for continuous operation.

1-4 Safety

1-4-1 Coil Operating Safety

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may potentially cause injury.

1-4-2 Patient Safety



WARNING!

Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.

1-4-3 Equipment Safety

All personnel using the coil should be instructed in the proper connection and handling of the coil.

Personnel should observe all warnings and cautions that appear in this guide.



WARNING!

Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.



WARNING!

The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.



CAUTION



Prevent cables from forming loops. Looping will degrade scan performance of the coil through RF coupling.

1-4-4 Electrical and Mechanical Safety



WARNING!

The coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it to ensure there is no external damage. Do not use the coil if the housing or cable is broken.

1-4-5 Modifications

Any changes or modifications to the coil must be approved and performed by USA Instruments prior to installation.

1-4-6 Reporting of Incidents

Users should contact USA Instruments immediately to report an incident and/or injury to a patient, operator or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

1-5 Coil Description

The 8-Channel Body Array is a receive-only coil designed to give optimum signal to noise ratio and uniform coverage of the torso and pelvis regions, including the chest, abdomen and hip anatomy. The coil is housed in a soft, flexible material that conforms to the patient anatomy to accommodate various body contours, while minimizing patient discomfort.

The two-piece coil design enables the clinical user to precisely position the coil over the region of interest for optimal results and also ensures that the coil can be used for imaging larger patients without suffering inconvenient signal voids.



CAUTION

United States federal law restricts this device to sale by, or on the order of, a physician.



Cables bearing this Coil ID label are compatible with software versions 10x and newer. With software versions 11x and newer, the system will generate an error when an incompatible coil is prescribed.

1-6 Coil Mode Configurations

To optimize image quality using the 8-Channel Body Array, focus on the coil configuration, patient positioning and FOV selection. Improper use of the coil is the major cause of image artifacts. The coil has a total of 12 elements, according to the coil configuration file selected.

8-Ch Body Upper: 8 coil elements are active for this mode, 4 posterior and 4 anterior (see Figure 1).

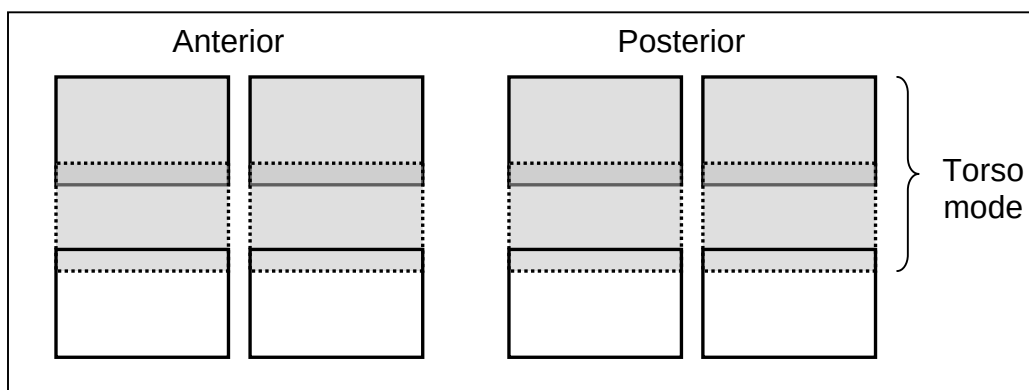


Figure 1: 8-Channel Body Upper active elements.

8-Ch Body Lower: 8 coil elements are active for this mode, 4 posterior and 4 anterior (see Figure 2).

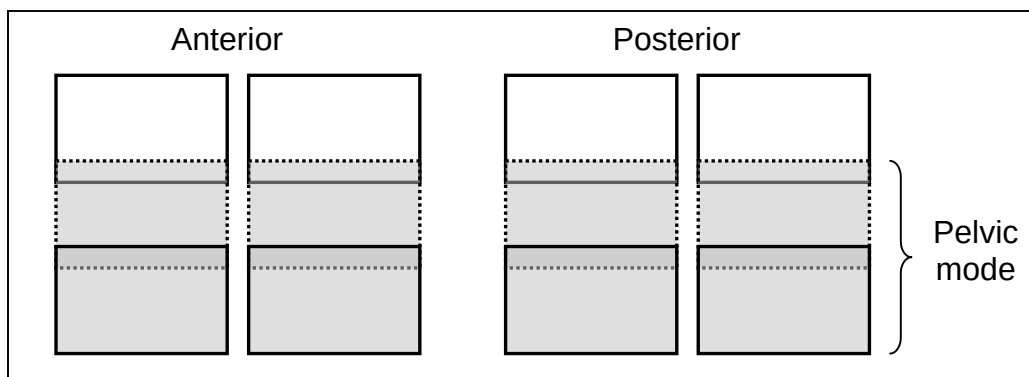


Figure 2: 8-Channel Body Lower active elements.

8-Ch Body FullFOV: 12 coil elements are active for this mode, 6 posterior and 6 anterior (see Figure 3).

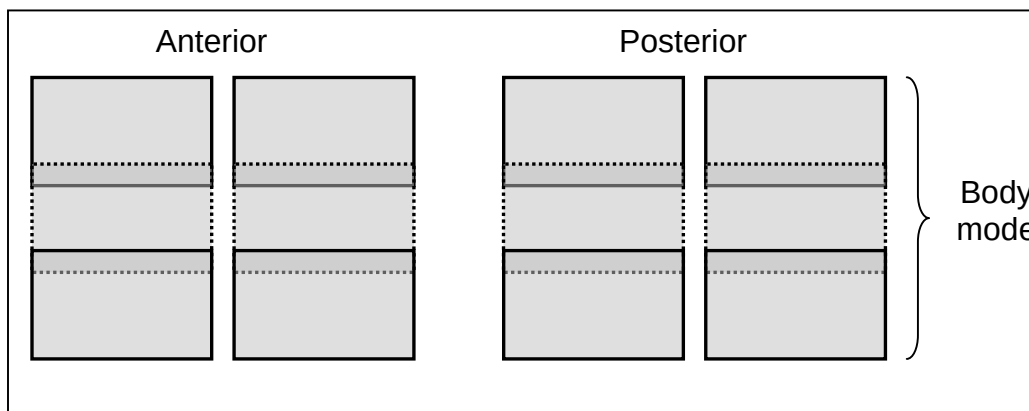


Figure 3: 8-Channel Body FullFOV active elements.

Element Dimensions

Right-Left direction: 9.4" (24.0 cm)

S-I direction: 7.1" (18.0 cm)

MAXIMUM RECOMMENDED FOV – TABLE 1-6

Modes	FOV
8Ch Body Upper	33cm
8Ch Body Lower	33cm
8Ch Body FullFOV	48cm

The coil is optimized for ASSET applications in all directions.

1-7 Environmental Requirements

ENVIRONMENTAL REQUIREMENTS – TABLE 1-7

	Transport and Storage	Operating Conditions
Barometric Pressure	645mm Hg to 795mm Hg	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	15° C to 35° C (59° F to 95° F)

Dimensions

Anterior: 21.0" L x 27.9" W (53.5cm x 71.1cm)

Posterior: 21.0" L x 30.5" W (53.5cm x 77.4cm)

Weight

Anterior: 4.4 lb (2.0 kg)

Posterior: 8.4 lb (3.8 kg)

1-8 Coil Specifications

COIL SPECIFICATIONS – TABLE 1-8

Field Strength	Coil Model	USAI Part Number	Frequency
1.5T	Prima 9000	100393	63.86 MHz

1-9 Coil Installation and Service

1-9-1 Coil Installation Procedure

Coil installation requires the services of your GE Service Representative.

1-9-2 Coil Service

The coil cannot be serviced on site. Contact your authorized GE Service Representative for service and maintenance needs.

SECTION 2 – SETUP

2-1 Positioning the Coil



WARNING!

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Step 1 – Place the posterior section of the Body Array onto the cradle along with the patient comfort pads (see *Figure 4*).

Step 2 – Place a clean cotton sheet over the coil so that the patient's skin will not come into direct contact with the coil.



Figure 4: Positioning the posterior section on the scanner cradle.

2-2 Positioning the Patient

Step 1 – Position the patient feet first, supine, with the region of interest lying on the posterior section of the coil (see *Figure 5*).

Step 2 – Position the anterior section above the region of interest. Reconnect the cable attaching the anterior and posterior sections of the coil. The cables should be positioned between the patient's legs.



Figure 5: Positioning the patient on the coil.



Figure 6: Attaching the connector to the anterior section.



Figure 7: Positioning the anterior section on the patient.

Step 3 – Fasten the Velcro straps for a snug and comfortable fit (see *Figure 8*). The center of the coil is indicated by cross marks on the anterior and posterior sections of the coil.

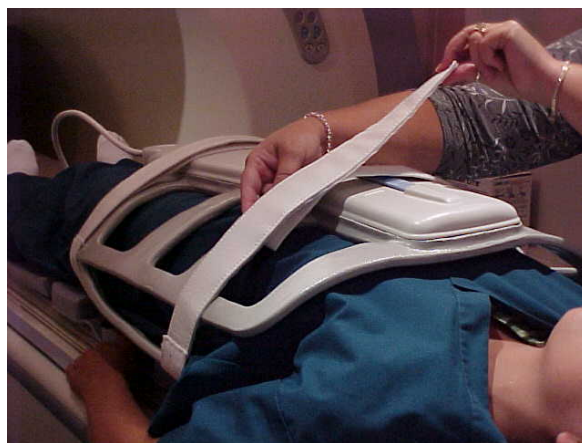


Figure 8: Fastening the Velcro straps.

2-3 Localizing Anatomy

Center the coil elements corresponding to the coil mode chosen over the region of interest. The landmark lines on the coil indicate the center of each coil mode. Refer to Section 1-6 for detailed descriptions.

2-4 Choosing the Correct Coil Mode Configuration

Refer to Section 1-6 for detailed descriptions of each coil mode. Choose the coil mode most appropriate for the corresponding anatomy of interest and required FOV. The Standard Body coil may be prescribed for localizing purposes or the 8Ch Body Upper, 8Ch Body Lower or 8Ch Body FullFOV coil modes can be used.

2-5 Recommended Scan Protocols

The coil is designed to be compatible with recommended protocols suggested by the system's manufacturer.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2377425-3	150480	1

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup and Test Execution for 11.0 M3A and Greater Software Versions

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position the phantom set on the coil without the pad, lining up the pegs on the phantom to the holes in the posterior section of the coil (see *Figure 9*).



Figure 9: Positioning the phantom on the coil.

3. Using the raised notches on top of the phantom, align the anterior section on the phantom and fasten the Velcro straps.

4. Landmark the coil at the indicated line and advance to scan (see *Figure 10*).



Figure 10: Landmarking the coil.

5. From the Common Service Desktop, select **Image Quality** and **Coil SNR**, then [Click here to start this tool.](#)
6. Select the coil type you want to run, enter the serial number and select **Start Scan**.
Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.

SNR Data Sheet

Use the table provided below to record the pass/fail results of the SNR test obtained from the Functional Checks section.

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

SECTION 4 – MAINTENANCE

4-1 Coil Care



WARNING!

Detach coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



CAUTION

Do not spray or pour the cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning the coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

4-3 Replaceable Accessories

The table below lists coil accessories that may be purchased for replacement.

REPLACEABLE ACCESSORIES LIST – TABLE 4-3

Part Name	GE Part #	USAI Part #
Coil	2377425-2	165142
Cable Assembly	2377425-7	155167
Phantom Kit (Phantom, Annular Rings, Positioners)	2377425-3	155168
Pad Kit (Patient Comfort, Ramp, Filler, Posterior)	2377425-6	155195
Anterior Handle Kit	2418437	N/A

For additional information, contact:

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USA800-558-5102
Canada.....800-668-0732

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India.....91-80-845-2923

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www.gemedicalsystems.com

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www.usainstruments.com

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	10/03	Teresa DeMarco	First Issue
02	03/06	Robert Brink	Revised figure10 to more clearly depict where to landmark coil